(+36) 204255258
Budapest, Hungary
ahmed.lagha@inf.elte.hu

Ahmed Fouad Lagha

PhD Researcher

ahmed-fouad-lagha.github.io
github.com/ahmed-fouad-lagha
linkedin.com/in/ahmed-fouad-lagha

PhD candidate in privacy-preserving ML with hands-on experience building ML models, LLM systems, and production web applications. Looking for a data science internship to apply strong analytical, engineering, and problem-solving skills to real-world data challenges.

Technical Skills

Languages & Tools	Python, SQL, C/C++, Java, Bash, Docker, Git, Linux, CI/CD, Flask, REST APIs
ML & Data Science	PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, SciPy
Privacy-Preserving ML	Federated Learning, Differential Privacy (DP-SGD)
Databases	MySQL, SQLite, MongoDB

Experience

PhD Researcher — Privacy-Preserving ML **Sep. 2025 – Present**
Eötvös Loránd University (ELTE) *Budapest, Hungary*

- Designing federated learning pipelines with differential privacy (DP-SGD) for synthetic data generation, achieving $\varepsilon = 1$ privacy guarantees while preserving model utility
- Building reusable evaluation pipelines to benchmark privacy–utility tradeoffs across models and datasets, enabling reproducible research
- Developing and delivering a lab on synthetic data generation with differential privacy for the Data Security course; mentoring 2 graduate students on DP-SGD and adversarial ML
- Awarded the Stipendium Hungaricum Scholarship — a competitive Hungarian government grant for academic excellence and research potential

Software Engineer Intern **Jun. 2023 – Feb. 2025**
ALX Africa *Remote*

- Built and shipped full-stack applications using Python, Flask, and SQL, delivering RESTful APIs for production use
- Designed normalized database schemas and optimized SQL queries, improving storage efficiency and cutting response times
- Collaborated within a distributed engineering team using Git, code reviews, and agile methodologies

Projects

MyVellum **Dec. 2025 – Present**
LLM Orchestration, RAG, Docker, CI/CD

- Architected an LLM orchestration system with modular prompt templates for an AI-powered autobiography platform serving 1,000+ users
- Developed vector-backed RAG pipelines with context-stitching, reducing hallucination rates in generated responses
- Integrated automated evaluation gates into CI/CD for relevance and safety metrics, ensuring quality across deployments
- Owned end-to-end API design and Docker deployment; mentored two engineers on prompt engineering best practices

WAU-Net **Jan. 2025 – Jun. 2025**
Master's Thesis — Python, PyTorch, OpenCV

- Designed a custom deep learning architecture for hair segmentation, achieving a 95.37% Dice coefficient on the benchmark dataset
- Engineered a data augmentation pipeline that improved generalization across diverse image conditions, reducing overfitting by 15%
- Evaluated model performance with precision, recall, and F1-score; optimized inference pipeline for real-time deployment

Education

PhD in Informatics — Privacy-Preserving Machine Learning **Sep. 2025 – Expected 2029**
Eötvös Loránd University (ELTE) *Budapest, Hungary*

M.Sc. in Artificial Intelligence **Sep. 2023 – Jun. 2025**
University of Khenchela *Khenchela, Algeria*

B.Sc. in Mathematics and Computer Science **Sep. 2020 – Jun. 2023**
University of Khenchela *Khenchela, Algeria*

Languages

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)